

# Instructions

This mock mirrors the BI3 flow: aptitude + reasoning, then DE technical MCQs, then SQL and coding.  
This is Mock 2 of 3; each uses a different question set.

## Section 1 - Aptitude

**Q1. A shirt costs Rs. 800, sold at 10% discount. Selling price:**

(A) Rs. 700

**(B) Rs. 720 ✓**

(C) Rs. 750

(D) Rs. 680

**Answer: (B) Rs. 720**

*Authenticity: Community Reported*

**Q2.  $\frac{3}{4}$  as a percentage:**

(A) 70%

**(B) 75% ✓**

(C) 80%

(D) 60%

**Answer: (B) 75%**

*Authenticity: Community Reported*

**Q3. A car travels 150 km in 3 hours. Speed:**

(A) 40 km/h

**(B) 50 km/h ✓**

(C) 60 km/h

(D) 45 km/h

**Answer: (B) 50 km/h**

*Authenticity: Community Reported*

**Q4. A finishes in 12 days, B in 6 days. Together:**

(A) 3 days

**(B) 4 days ✓**

(C) 5 days

(D) 6 days

**Answer: (B) 4 days**

*Authenticity: Community Reported*

**Q5. A tap fills a tank in 5 hours. Fraction filled in 2 hours:**

(A)  $\frac{1}{5}$

**(B)  $\frac{2}{5}$  ✓**

(C)  $\frac{3}{5}$

(D)  $\frac{1}{2}$

**Answer: (B)  $\frac{2}{5}$**

*Authenticity: Community Reported*

Q6. 6 men do a job in 10 days. 12 men take:

(A) 5 days ✓

(B) 6 days

(C) 8 days

(D) 4 days

Answer: (A) 5 days

Authenticity: Research-Backed Company Style

## Section 2 - Reasoning

Q1. Next term: 2, 4, 8, 16, ?

(A) 24

(B) 30

(C) 32 ✓

(D) 20

Answer: (C) 32

Authenticity: Frequently Repeated

Q2. Next term: 1, 4, 9, 16, ?

(A) 20

(B) 25 ✓

(C) 24

(D) 36

Answer: (B) 25

Authenticity: Frequently Repeated

Q3. If CAT = DBU (each +1), then DOG is:

(A) EPH ✓

(B) EPG

(C) FPH

(D) DPG

Answer: (A) EPH

Authenticity: Community Reported

Q4. All engineers are graduates. Some graduates are managers. So some engineers are managers.

(A) Follows

(B) Does not follow ✓

(C) Always true

(D) None

Answer: (B) Does not follow

Authenticity: Frequently Repeated

Q5. Class of 40; student ranks 12th from top. Rank from bottom:

(A) 28th

**(B) 29th ✓**

(C) 30th

(D) 27th

**Answer: (B) 29th**

*Authenticity: Community Reported*

## Section 3 - Data Engineering Technical

Q1. Which JOIN keeps all left rows plus matches (NULLs otherwise)?

(A) INNER JOIN

**(B) LEFT JOIN ✓**

(C) CROSS JOIN

(D) SELF JOIN

**Answer: (B) LEFT JOIN**

*Authenticity: Community Reported*

Q2. Which quickly removes all rows and cannot be easily rolled back?

(A) DELETE

**(B) TRUNCATE ✓**

(C) DROP

(D) REMOVE

**Answer: (B) TRUNCATE**

*Authenticity: Community Reported*

Q3. A central fact table with denormalized dimensions is a:

(A) Snowflake schema

**(B) Star schema ✓**

(C) Galaxy schema

(D) Network schema

**Answer: (B) Star schema**

*Authenticity: Community Reported*

Q4. SCD Type 2 handles a change by:

(A) Overwriting old value

**(B) Adding a new versioned row ✓**

(C) Deleting the record

(D) Ignoring it

**Answer: (B) Adding a new versioned row**

*Authenticity: Community Reported*

**Q5. In ELT, transformation happens:**

(A) Before extraction

**(B) After loading into the target ✓**

(C) Never

(D) During extraction only

**Answer: (B) After loading into the target**

*Authenticity: Community Reported*

**Q6. Idempotency means re-running a step:**

(A) Doubles data

**(B) Produces the same result without duplicates ✓**

(C) Fails

(D) Deletes source

**Answer: (B) Produces the same result without duplicates**

*Authenticity: Community Reported*

**Q7. The Snowflake compute resource that runs queries is a:**

(A) Stage

**(B) Virtual warehouse ✓**

(C) Micro-partition

(D) Role

**Answer: (B) Virtual warehouse**

*Authenticity: Community Reported*

**Q8. Amazon S3 / Azure Blob are best described as:**

(A) Relational DBs

**(B) Scalable object/file storage ✓**

(C) Message queues

(D) Compute clusters

**Answer: (B) Scalable object/file storage**

*Authenticity: Community Reported*

**Q9. pandas method to drop rows with NaN:**

(A) fillna()

**(B) dropna() ✓**

(C) isnull()

(D) merge()

**Answer: (B) dropna()**

*Authenticity: Community Reported*

**Q10. groupby() is usually followed by:**

(A) A sort only

**(B) An aggregation like sum()/mean() ✓**

(C) Deletion

(D) Nothing

**Answer: (B) An aggregation like sum()/mean()**

*Authenticity: Community Reported*

## Section 4 - SQL

### Problem: Nth Highest Salary (LIMIT/OFFSET)

Find the 3rd highest distinct salary using ORDER BY with LIMIT/OFFSET.

```
SELECT DISTINCT salary
FROM emp
ORDER BY salary DESC
LIMIT 1 OFFSET 2;
```

#### Expected Output

```
salary
200
```

## Section 5 - Coding / Python

### Problem 1: Contains Duplicate

Difficulty: **Easy** | Pattern: **Set** | Authenticity: **Community Reported**

#### Problem Statement

Return True if any value appears at least twice.

#### Input Format

Line 1: n. Line 2: n integers.

#### Output Format

True or False.

#### Constraints

$1 \leq n \leq 10^5$

#### Sample Input

```
5
1 2 3 1 4
```

#### Sample Output

```
True
```

#### Python Solution

```
def contains_duplicate(nums):
    return len(set(nums)) != len(nums)

print(contains_duplicate([1, 2, 3, 1, 4]))
```

#### Step-by-Step Explanation

A set drops duplicates; a smaller set size means a duplicate existed.

#### Dry Run

```
set size 4 != list size 5 -> True
```

Time Complexity: **O(N)** | Space Complexity: **O(N)**

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## Problem 2: Factorial

Difficulty: *Easy* | Pattern: *Iteration* | Authenticity: *Community Reported*

### Problem Statement

Compute  $n!$  for a non-negative integer  $n$ .

### Input Format

A single integer  $n$ .

### Output Format

$n!$ .

### Constraints

$0 \leq n \leq 20$

### Sample Input

5

### Sample Output

120

### Python Solution

```
def factorial(n):  
    result = 1  
    for i in range(2, n + 1):  
        result *= i  
    return result  
  
print(factorial(5))
```

### Step-by-Step Explanation

Multiply  $2..n$ ; empty loop handles  $0! = 1! = 1$ .  $5! = 120$ .

### Dry Run

$1*2*3*4*5=120$

Time Complexity: **O(N)** | Space Complexity: **O(1)**

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# BI3 Technologies

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# Full-Length Mock Test 2

Applied Role(s):

Data Engineer - Trainee

Online Assessment Preparation Material

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Every question carries an authenticity label. See the legend on the next page.

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## Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

| Label                                | Meaning                                                                                              |
|--------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Verified Previous-Year</b>        | The specific problem/pattern is documented from actual reported assessment experiences.              |
| <b>Frequently Repeated</b>           | The topic or pattern type is documented as recurring across multiple hiring cycles.                  |
| <b>Community Reported</b>            | A widely shared, role-relevant pattern reported across placement communities and forums.             |
| <b>Research-Backed Company Style</b> | Authored to match the researched difficulty, style and pattern - practice, not a real past question. |

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.